SECTION 1

Linear Regression

- **Model form:** Predictions are a weighted sum of features plus bias; vector notation bundles parameters in θ for compact math and code.
- **Training objective:** Minimizing MSE (often reported as RMSE) finds θ that best fits labeled training points.
- **Normal equation:** For moderate feature counts, a closed-form least-squares solution (via pseudoinverse/SVD) solves for θ directly.
- **Complexity trade-offs:** Exact methods scale poorly with very large feature dimensionality, while prediction after training is cheap and linear in instance size.

SECTION 2

Gradient Descent

- **Idea:** Iteratively move θ along the negative gradient of the cost with step size controlled by the learning rate η.
- **Batch GD:** Each update uses the full training set's gradient, stable for convex linear least squares but slow on huge data.
- **Stochastic and mini-batch:** Single-instance or small-batch updates add noise but enable speed, online learning, and escape from some local roughness in non-convex tasks.
- **Practical needs:** Feature scaling reduces ill-conditioned valleys; learning schedules and tolerances govern convergence; `SGDRegressor` mirrors these ideas in Scikit-Learn.

SECTION 3

Polynomial Regression

- **Linear in parameters, nonlinear in inputs:** Adding powers (and cross terms) lets a linear solver fit curved relationships.
- **Combinatorial cost:** High polynomial degree explodes feature count, risking slow training and severe overfitting.
- **Model selection angle:** Degree choice balances flexibility vs. stability on the same underlying algorithmic family.

SECTION 4

Learning Curves

- **Diagnostic plots:** Training and validation error vs. training set size separate underfitting from overfitting without guessing from a single score.
- **Underfitting signal:** Both curves plateau high and close together—more data alone rarely fixes a too-simple hypothesis.
- **Overfitting signal:** Low training error with higher validation error and a gap suggests variance; more data or simpler models may help.
- **Bias–variance view:** Error decomposes conceptually into wrong assumptions, sensitivity to sampling noise, and irreducible label noise.

SECTION 5

Regularized Linear Models

- **Ridge (■2):** Shrinks weights toward zero with α-weighted squared penalties, stabilizing coefficients when features correlate or noise is high.
- **Lasso (■1):** Encourages sparse solutions by driving some weights exactly to zero, performing implicit feature selection.
- **Elastic Net:** Mixes ■1 and ■2 penalties to keep sparsity while handling correlated groups better than pure lasso in some regimes.
- **Early stopping:** Halting iterative training when validation error bottoms out regularizes without an explicit penalty term.

SECTION 6

Logistic Regression and Softmax Regression

- **Binary probabilities:** A linear score passed through the logistic sigmoid yields class probabilities; a 0.5 threshold on that probability sets the class decision boundary.
- **Training loss:** Log loss penalizes confident wrong predictions; convexity enables reliable gradient-based fitting (no normal equation).
- **Multiclass softmax:** One weight vector per class produces scores normalized by softmax into a probability simplex; prediction picks the argmax class for mutually exclusive labels.
- **Regularization note:** Scikit-Learn's `LogisticRegression` uses inverse-strength C for ■2 by default; `multi_class="multinomial"` engages softmax training with compatible solvers.